

# LABORATORY OBSERVATION

**Course:** Full Stack Web Development

**Course Code:** 23CM4121

**Observation Task No.:** Task2

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**Roll No.:** A24126552122

**Date.:**

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## 1. TITLE

My Styled Webpage — HTML5 Graphics and CSS3 Web Styling

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## 2. AIM

To understand how CSS3 enhances the appearance of web pages and to explore HTML5 graphics using the Canvas element.

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## 3. OBJECTIVE

- To draw basic shapes and text on a webpage using the HTML5 canvas element.
  - To understand CSS3 syntax and selectors used to style HTML elements.
  - To apply colors, fonts, borders, margins, padding, and backgrounds to a webpage.
  - To use box-shadow, text-shadow, and border-radius for visual enhancement.
  - To apply a gradient background and hover effects on navigation links and buttons.
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## 4. THEORY

The HTML5 <canvas> element provides a blank drawing surface on which shapes, lines, and text can be drawn using JavaScript. Drawing is done through a 2D rendering context obtained with `getContext("2d")`, which provides methods such as `fillRect()` for rectangles, `arc()` for circles, `moveTo()/lineTo()` with `stroke()` for lines, and `fillText()` for text.

CSS3 is used to style the visual appearance of a webpage through selectors that target elements, classes, or IDs. Properties such as color, font-family, border, margin, and padding control basic appearance, while background and linear-gradient() add colour or gradient fills. Effects such as box-shadow and text-shadow add depth around elements and text, and border-radius rounds element corners. The :hover pseudo-class applies a style only when the mouse is over an element, which is used to create interactive effects on navigation links and buttons.

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## 5. ALGORITHM / PROCEDURE

1. Create a new HTML file and declare <!DOCTYPE html> at the top.
2. Add a <style> section inside <head> for CSS3 styling.
3. Create a <header> with a heading.
4. Create a <nav> bar with menu links.

5. Add a <canvas> element for drawing shapes.
  6. Get the canvas element and its 2D context using JavaScript.
  7. Draw a rectangle using fillRect().
  8. Draw a circle using arc(), fill(), and stroke().
  9. Draw a line using moveTo(), lineTo(), and stroke().
  10. Display the student's name on the canvas using fillText().
  11. Create three content cards using <div class="card">, each with a heading, text, and button.
  12. Style the header, nav, and cards using CSS3 properties such as gradient, shadow, and border-radius.
  13. Add hover effects on the nav links and card buttons.
  14. Add a <footer> with copyright text.
  15. Save the file and open it in a browser to check the output.
  16. Test the canvas drawing and hover effects, and record the results as test cases.
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## 6. TASK DESCRIPTION

A webpage titled "My Styled Webpage" was created, covering the following parts:

### Part A: HTML5 Canvas

Using the <canvas> element, drew:

- One rectangle
- One circle
- One line
- The student's name as text

### Part B: CSS3 Styling

Designed the webpage with the following components:

- Header
- Navigation Bar
- Three Content Cards
- Footer

Applied the following CSS properties:

- Background colors and gradient background
  - Different font families and sizes
  - Borders and border radius
  - Margin and padding
  - Box shadow and text shadow
  - Hover effect for buttons
  - Hover effect for navigation links
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## 7. PROGRAM

### Task2.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
```

```
<title>My Styled Webpage</title>
<style>
body {
  margin: 0;
  padding: 0;
  font-family: Arial, sans-serif;
}
/* Header */
header {
  background: linear-gradient(to right, #4facfe, #00f2fe);
  color: white;
  text-align: center;
  padding: 25px;
  text-shadow: 5px 2px 4px rgba(0,0,0,0.4);
}
header h1 {
  font-size: 32px;
  font-family: 'Trebuchet MS', sans-serif;
}
/* Navigation Bar */
nav {
  background-color: #333;
  padding: 12px;
  text-align: center;
}
nav a {
  color: white;
  text-decoration: none;
  margin: 0 15px;
  font-size: 16px;
  font-family: Verdana, sans-serif;
}
nav a:hover {
  color: #00f2fe;
  text-shadow: 1px 1px 3px black;
}
/* Canvas Section */
.canvas-section {
  text-align: center;
  padding: 20px;
}
canvas {
  border: 1px solid #000;
  background-color: white;
}
/* Content Cards */
.card-container {
  display: flex;
  justify-content: center;
  gap: 20px;
```

```

padding: 30px;
flex-wrap: wrap;
}
.card {
background-color: #f9f9f9;
border: 1px solid #ccc;
border-radius: 12px;
padding: 20px;
width: 220px;
box-shadow: 4px 4px 12px rgba(0,0,0,0.2);
text-align: center;
}
.card h3 {
font-family: Georgia, serif;
color: #333;
}
.card button {
margin-top: 10px;
padding: 8px 16px;
border: none;
border-radius: 6px;
background-color: #4facfe;
color: white;
cursor: pointer;
}
.card button:hover {
background-color: #00c6ff;
box-shadow: 2px 2px 8px rgba(0,0,0,0.3);
}
/* Footer */
footer {
background-color: #333;
color: white;
text-align: center;
padding: 15px;
margin-top: 20px;
}
</style>
</head>
<body>

<!-- Header -->
<header>
  <h1>My Styled Webpage</h1>
</header>

<!-- Navigation Bar -->
<nav>
  <a href="#">Home</a>
  <a href="#">About</a>

```

```

<a href="#">Services</a>
<a href="#">Contact</a>
</nav>

<!-- ===== PART A: HTML5 Canvas ===== -->
<div class="canvas-section">
  <h2>Part A - HTML5 Canvas</h2>
  <canvas id="myCanvas" width="400" height="300"></canvas>
</div>

<script>
  const canvas = document.getElementById("myCanvas");
  const ctx = canvas.getContext("2d");

  // 1. Draw a Rectangle
  ctx.fillStyle = "skyblue";
  ctx.fillRect(30, 30, 120, 80);

  // 2. Draw a Circle
  ctx.beginPath();
  ctx.arc(280, 70, 50, 0, 2 * Math.PI);
  ctx.fillStyle = "orange";
  ctx.fill();
  ctx.stroke();

  // 3. Draw a Line
  ctx.beginPath();
  ctx.moveTo(30, 180);
  ctx.lineTo(350, 180);
  ctx.strokeStyle = "black";
  ctx.lineWidth = 3;
  ctx.stroke();

  // 4. Display Name as Text
  ctx.font = "24px Arial";
  ctx.fillStyle = "darkblue";
  ctx.fillText("Akshaya", 130, 250);
</script>

<!-- ===== PART B: CSS3 Styling ===== -->
<h2 style="text-align:center;">Part B - CSS3 Styling</h2>
<div class="card-container">
  <div class="card">
    <h3>Card One</h3>
    <p>This is the first content card with styling.</p>
    <button>Learn More</button>
  </div>
  <div class="card">
    <h3>Card Two</h3>
    <p>This is the second content card with styling.</p>
  </div>
</div>

```

```

    <button>Learn More</button>
  </div>
  <div class="card">
    <h3>Card Three</h3>
    <p>This is the third content card with styling.</p>
    <button>Learn More</button>
  </div>
</div>

<!-- Footer -->
<footer>
  <p>&copy; 2026 My Styled Webpage. All Rights Reserved.</p>
</footer>

</body>
</html>

```

## 8. CODE EXPLANATION

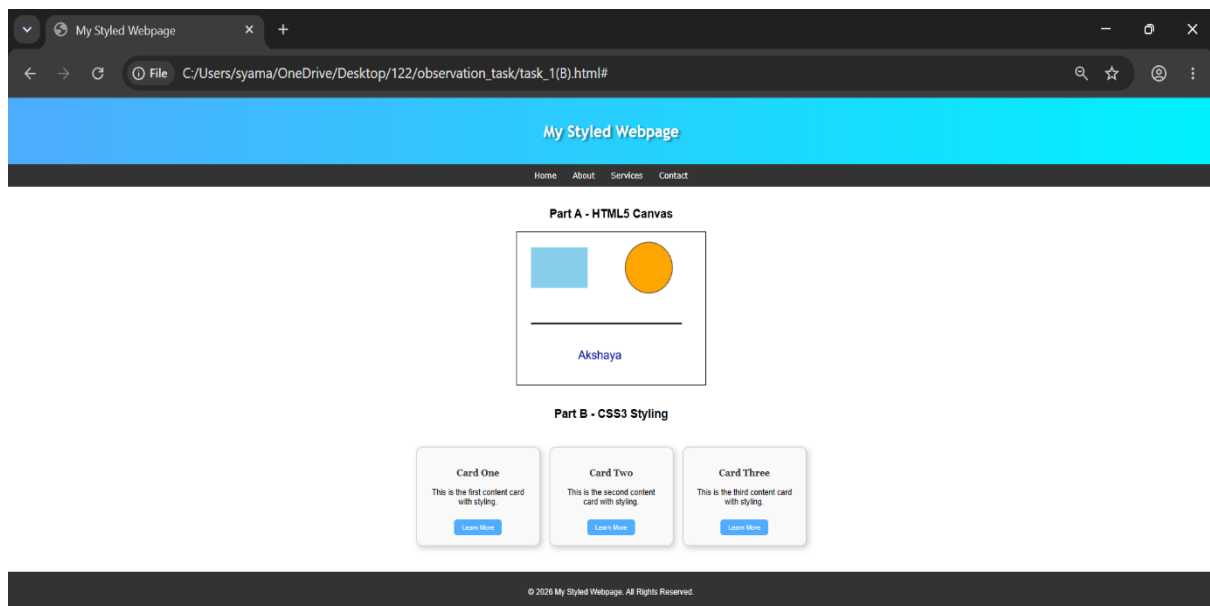
| Tag / Function      | Purpose                                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------|
| <canvas>            | Creates a drawing area on the page where shapes and text can be drawn using JavaScript.                  |
| <script>            | Contains the JavaScript code used to draw on the canvas.                                                 |
| getElementById()    | Selects an HTML element (the canvas) by its id so it can be used in JavaScript.                          |
| getContext("2d")    | Gets the 2D drawing context of the canvas, which provides all the drawing functions.                     |
| fillRect()          | Draws a filled rectangle at a given position and size.                                                   |
| beginPath()         | Starts a new drawing path so each new shape is drawn separately.                                         |
| arc()               | Draws a circular path, used here to create the circle.                                                   |
| fill()              | Fills a shape with the selected colour.                                                                  |
| stroke()            | Draws the outline of a shape or draws a line.                                                            |
| moveTo() / lineTo() | Sets the starting point and ending point of a straight line.                                             |
| fillText()          | Displays text (the student's name) at a given position on the canvas.                                    |
| <header>            | Defines the top section of the page containing the main heading.                                         |
| <nav>               | Defines the navigation bar containing menu links.                                                        |
| <div class="card">  | Creates a content card box used to display a heading, text, and button.                                  |
| <button>            | Creates a clickable button, used inside each card.                                                       |
| <footer>            | Defines the bottom section of the page containing copyright text.                                        |
| linear-gradient()   | Creates a smooth colour transition, used as the header's background.                                     |
| box-shadow          | Adds a shadow around an element, used on the cards and buttons.                                          |
| text-shadow         | Adds a shadow behind text, used on the header and nav links.                                             |
| border-radius       | Rounds the corners of the cards and buttons.                                                             |
| :hover              | Pseudo-class that applies a style only when the mouse is over an element, used on nav links and buttons. |

## 9. TEST CASES AND EXECUTION DATA

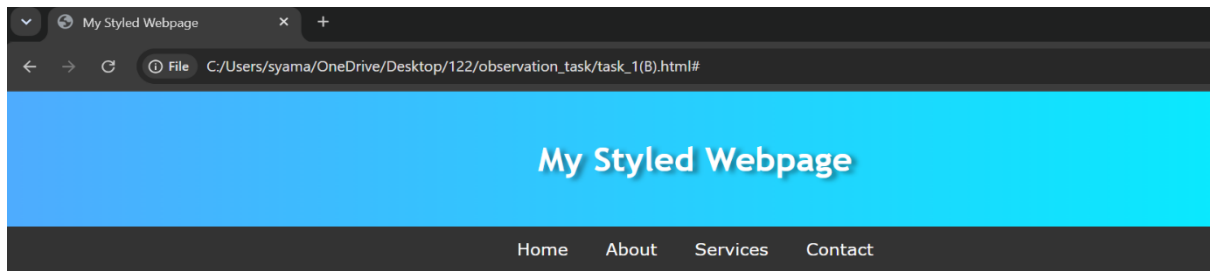
| Test Case | Action                                  | Expected Result                                    | Actual Result                       | Status |
|-----------|-----------------------------------------|----------------------------------------------------|-------------------------------------|--------|
| TC-01     | Open the HTML file                      | Header, nav, canvas, cards, and footer all display | Displayed correctly                 | Pass   |
| TC-02     | Check the canvas drawing                | Rectangle, circle, line, and name text appear      | All shapes and text drawn correctly | Pass   |
| TC-03     | Hover over a nav link                   | Link colour changes with a text shadow             | Colour changed as expected          | Pass   |
| TC-04     | Hover over a card's "Learn More" button | Button background and shadow change                | Changed correctly on hover          | Pass   |
| TC-05     | Check the content cards                 | Cards show rounded corners and a box shadow        | Rounded corners and shadow visible  | Pass   |

## 10. OUTPUT

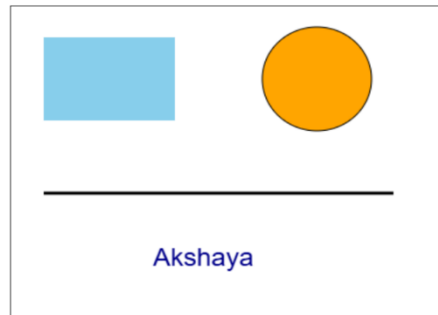
### Output 1:



### Output 2: Canvas Drawing

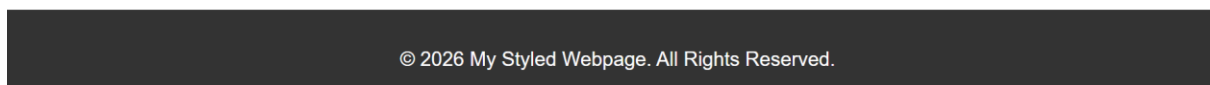
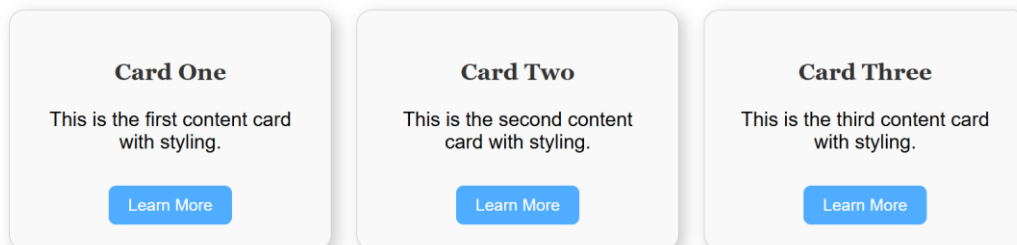


### Part A - HTML5 Canvas

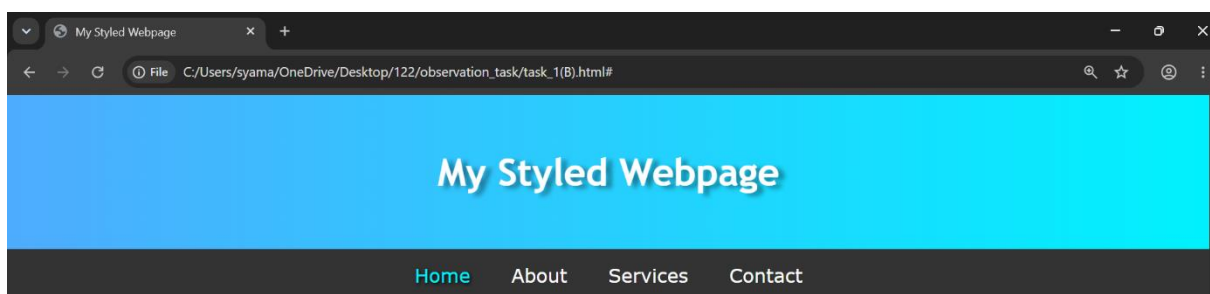


## Output 3: Content Cards

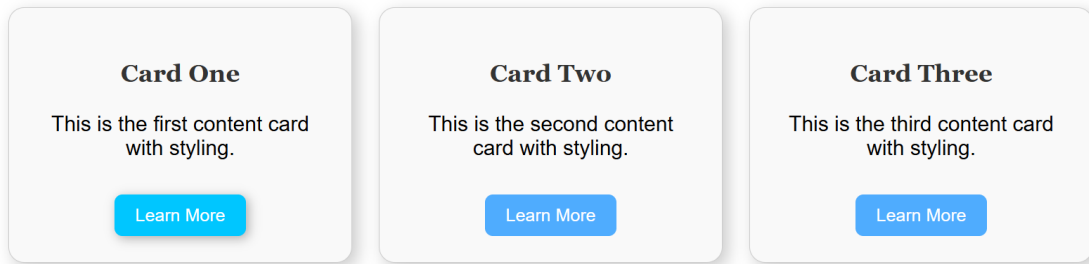
### Part B - CSS3 Styling



## Output 4: Hover Effects (Nav Link and Button)



## Part B - CSS3 Styling



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## 11. RESULT

Thus, a webpage titled "My Styled Webpage" was successfully developed using HTML5 Canvas and CSS3. Basic shapes and text were drawn on the canvas, and the page was styled with gradients, shadows, rounded corners, and hover effects. The page was tested in a web browser and all features were verified to work correctly.